Homework #5
Database Systems
(75 Points)

(No late submission accepted -- Submit to Brightspace by Midnight on May 07, 2025)

Please remember to include the following statement at the beginning of your submitted assignment and SIGN it. Your assignment won't be graded with the signed statement.

"I have done this assignment completely on my own. I have not copied it, nor have I given my solution to anyone else. I understand that if I am involved in plagiarism or cheating, I will have to sign an official form that I have cheated and that this form will be stored in my official university record. I also understand that I will receive a grade of 0 for the involved assignment and my grade will be reduced by one level (e.g., from A to A- or from B+ to B) for my first offense, and that I will receive a grade of "F" for the course for any additional offense of any kind."

1. As we discussed Object-Oriented PL/SQL allows defining an object type, which helps in designing object-oriented database in Oracle. An object type allows you to create composite types. Using objects allows you to implement real world objects with specific structure of data and methods for operating it. Objects have attributes and methods. Attributes are properties of an object and are used for storing an object's state; and methods are used for modeling its behavior. Member methods are used for manipulating the attributes of the object. The Map method is a function implemented in such a way that its value depends upon the value of the attributes. Submit .sql file(s) for the following. (15+15+15)

   i. Create an object '*quadrangle*' with parameters length and width, member function '*enlarge*', map member function '*measure*' and member procedure '*display*'.
   ii. Create a Body '*quadrangle*' to implement your logic for '*enlarge'*, '*measure*' and '*display*'. As the name suggested '*enlarge*' append/add values in the parameters, '*measure*' should be computed using the formula sqrt(L^2+W^2) and '*display*' should show all the values on screen after execution.
   iii. Call functions/procedures with parameters, display results and show at least a comparison with IF-ELSE condition.

   https://www.tutorialspoint.com/plsql/plsql_object_oriented.htm

2. Consider the join R $\infty_{R.A = S.B}$ S, where R and S are two relations. Three join methods, i.e., nested loop, sort merge and hash join, are discussed in the class. Nested loop and sort merge may benefit from the existence of indexes and/or whether or not the tables are sorted based on the joining attributes. Identify three different scenarios (i.e., with given sizes of R and S, the index status on R.A and/or S.B, and whether the tables are sorted based on R.A and/or S.B) such that each of the following claims is true for one of the scenarios.

   (a) (10 points) Nested loop outperforms sort merge and hash join.
   (b) (10 points) Sort merge outperforms nested loop and hash join.
   (c) (10 points) Hash join outperforms nested look and sort merge.

   Method 1 is said to "outperform" Method 2 if (1) Method 1 incurs a smaller number of I/O pages than Method 2, or (2) Method 1 and Method 2 incur the same number of I/O pages but Method 1 incurs less CPU cost (e.g., needs fewer comparisons) than Method 2. Justify your answer.